

ENANTIOMERS, DIASTEREOMERS, MESO COMPOUNDS AND RACEMIC MIXTURES

Point-wise Notes

1. Enantiomers

- Enantiomers** are stereoisomers that are non-superimposable mirror images of each other.
- For a molecule containing one stereogenic centre, the two opposite configurations form an enantiomeric pair; e.g. (R)-2-butanol and (S)-2-butanol.
- For molecules with several stereogenic centres, corresponding configurations are inverted at all stereogenic centres in a true enantiomeric pair.
- In an achiral environment, two pure enantiomers have identical ordinary physical properties.
 - Same melting point and boiling point.
 - Same density, vapour pressure and refractive index.
 - Same heat of combustion and other ordinary thermodynamic properties.
 - Same solubility in an achiral solvent.
- Their ordinary IR, UV-visible, mass and NMR spectra in an achiral medium are essentially identical.
- They differ in their interaction with plane-polarized light.
$$[\alpha](+) = -[\alpha](-)$$
- If one enantiomer rotates plane-polarized light clockwise by a certain magnitude, the other rotates it anticlockwise by the same magnitude under identical conditions.
- Dextrorotatory rotation is denoted by (+) or d; levorotatory rotation is denoted by (-) or l.
- There is no general relation between R/S configuration and the sign of optical rotation.

R does not imply (+) ; S does not imply (-)

- The sign and magnitude of optical rotation must be determined experimentally.
- In an achiral environment, enantiomers have the same energy and normally react at the same rate with an achiral reagent.
- With an achiral reagent, corresponding products formed from the two enantiomers are usually themselves enantiomerically related.
- Enantiomers may behave differently in a chiral environment because their interactions become diastereomeric and therefore need not have equal energies.
 - Chiral reagent or catalyst.
 - Chiral solvent.
 - Chiral stationary phase.
 - Enzyme, receptor or other biological macromolecule.
- Biological systems are chiral; therefore enantiomers may show different taste, smell, metabolism, pharmacological action or toxicity.
- Limonene example: (R)-limonene and (S)-limonene are enantiomers and are commonly cited as giving different odour perceptions because olfactory receptors are chiral.
- Histidine example: L-histidine and D-histidine are enantiomers. They have the same constitution but opposite three-dimensional configuration at the alpha-carbon.
- Ordinary distillation, crystallisation and chromatography generally cannot separate enantiomers efficiently because their properties are identical in an achiral environment.

- Resolution requires a chiral discrimination, such as a chiral stationary phase, an optically active resolving agent or an enzyme.

2. Diastereomers

- Diastereomers** are stereoisomers that are not mirror images of each other.
- For compounds containing several stereogenic centres, two stereoisomers are diastereomers when the configuration differs at one or more, but not all, corresponding stereogenic centres.
- Example: (R,R) and (R,S) are diastereomers; similarly, (R,R) and (S,R) are diastereomers.
- By contrast, (R,R) and (S,S) are generally enantiomers when no internal symmetry makes them identical.
- Geometrical isomers are an important class of diastereomers.
- cis-2-Butene and trans-2-butene are diastereomers.
- cis-1,2-Dichloroethene and trans-1,2-dichloroethene are also diastereomers.
- Diastereomers generally have different physical properties.
- Diastereomers may have different chemical properties even toward achiral reagents.
- Their rates, activation energies, equilibrium constants and product distributions may differ.
- Their optical rotations have no necessary relation in magnitude or sign.
- One diastereomer may be dextrorotatory, another levorotatory, or one may even be optically inactive.
- Because their physical properties differ, diastereomers can usually be separated by ordinary crystallisation, distillation or chromatography.
- An epimer is a special type of diastereomer that differs in configuration at only one of several stereogenic centres.
 - Melting point and boiling point may differ.
 - Density, refractive index and solubility may differ.
 - IR/NMR spectra may differ because the molecules are not related by mirror symmetry.

3. Meso Compounds

- A **meso compound** contains two or more stereogenic centres but the complete molecule is achiral.
- The presence of stereogenic centres therefore does not automatically imply optical activity.
- A meso compound contains an internal symmetry relationship that makes the molecule superimposable on its mirror image.
- A plane of symmetry is a common reason for meso behaviour, although molecular symmetry must be considered in the actual three-dimensional conformation.
- A meso compound is a single pure compound, not a mixture.
- It is optically inactive because optical effects are compensated within the same molecule.
- This is called internal compensation.
- Classical example: meso-tartaric acid.
- Another standard example: meso-2,3-dibromobutane.
- For a symmetrical two-centre system, (R,R) and (S,S) may form an enantiomeric pair, while the internally symmetric (R,S) form may be meso.
- However, R,S does not automatically mean meso; the required molecular symmetry must actually be present.
- A molecule can contain R and S stereogenic centres and still be chiral if the two halves are not related by symmetry.

13. Meso forms reduce the actual number of stereoisomers below the simple maximum 2^n .
14. The mirror image of a meso compound is not a distinct enantiomer because it is superimposable on the original molecule.
15. In tartaric acid, the two optically active forms constitute an enantiomeric pair, whereas meso-tartaric acid is achiral and optically inactive.

Meso optical inactivity = internal compensation

4. Racemic Mixture / Racemic Modification

1. A **racemic mixture** contains equal amounts of two enantiomers.
2. For a simple one-centre system, it may be written as 50% R + 50% S or as (+/-).
3. The equal and opposite optical rotations cancel, so the net observed rotation is zero.
4. The optical inactivity is due to cancellation between different molecules and is therefore called external compensation.
5. Each individual molecule in the racemate is still chiral and, when isolated as a pure enantiomer, is optically active.
6. A racemic mixture must not be confused with a meso compound: a racemate is a mixture of two chiral enantiomers, whereas a meso compound is one achiral substance.
7. Mixing equimolar amounts of two pure enantiomers gives a racemic mixture.
8. An achiral reaction system can produce a racemate when two enantiotopic pathways are equally probable.
9. Conversion of an optically active substance into its racemic form is called racemisation.
10. Racemisation commonly occurs when the stereogenic centre passes through an achiral or effectively planar intermediate and is regenerated from either face.
11. Enolisation is a classical route by which compounds containing an acidic stereogenic α -carbon can racemise.
12. Racemisation may also occur through reversible bond cleavage or other intermediates that destroy the original stereochemical information.

Racemate optical inactivity = external compensation

Solid-state forms of racemic modification

13. **Racemic mixture / conglomerate:** the two enantiomers crystallise as separate enantiomorphous crystals. The bulk sample contains equal amounts of both crystal types.
14. **Racemic compound:** the two enantiomers occur together in a definite 1:1 arrangement in the same crystal lattice, producing a distinct crystalline phase.
15. **Racemic solid solution / pseudoracemate:** the two enantiomers occupy lattice positions statistically or randomly in a common crystal lattice rather than forming a fixed 1:1 molecular compound.
16. The solid-state form is important because the melting point, solubility, density and phase behaviour of a racemate may differ from those of the pure enantiomers.
17. A racemic compound can have a melting point higher or lower than that of the pure enantiomer because its crystal lattice is a different phase.
18. A conglomerate consists of separate crystals of the two enantiomers; in suitable cases, such crystals can be resolved mechanically.

19. Some systems show a transition temperature at which the preferred racemic solid form changes from one modification to another.

How the nature of a racemic modification may be studied

20. Melting-point or freezing-point behaviour and composition-temperature curves.
21. Solubility behaviour and solubility-composition curves.
22. Crystal morphology when separate enantiomorphous crystals are formed.
23. X-ray diffraction or other solid-state structural methods.

Resolution of a racemic mixture

24. Resolution means separation of a racemate into its individual enantiomers.
25. Direct separation by ordinary achiral methods is difficult because the enantiomers have identical properties in an achiral environment.
26. A classical method converts the two enantiomers into diastereomeric salts or derivatives using an optically active resolving agent.
27. The resulting diastereomers have different solubilities and other physical properties and can therefore be separated, commonly by fractional crystallisation.
28. After separation, the resolving group is removed to regenerate the individual enantiomers.
29. Racemic acids can be resolved with a chiral base; racemic bases can similarly be resolved with a chiral acid.
30. Other methods include mechanical separation of enantiomorphous crystals, biochemical or enzymatic resolution, chiral chromatography, kinetic resolution and preferential crystallisation.

Meso compound and racemic mixture - essential distinction

- Meso compound: one achiral compound; optical inactivity arises from internal compensation.
- Racemic mixture: equal mixture of two chiral enantiomers; optical inactivity arises from external compensation.
- A meso compound cannot be resolved into enantiomers because it is itself achiral.
- A racemic mixture can be resolved because two enantiomeric components are actually present.